EDS 223: Homework Assignment 3

Investigating Potential West Coast Marine Aquaculture

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Load Packages

```
library(tidyverse)
library(terra)
library(sf)
library(tmap)
```

Prepare Data

```
# Read sea surface temperature (SST) data from 2008-2012
sst_2008 <- rast('data/average_annual_sst_2008.tif')
sst_2009 <- rast('data/average_annual_sst_2009.tif')
sst_2010 <- rast('data/average_annual_sst_2010.tif')
sst_2011 <- rast('data/average_annual_sst_2011.tif')
sst_2012 <- rast('data/average_annual_sst_2012.tif')

# Read bathymetry data
depth <- rast('data/depth.tif')

# Read exclusive economic zones (EEZ) data
eez <- st_read('data/wc_regions_clean.shp')

# Combine the SST(2008-2012) rasters into a raster stack
sst <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)
```

```
# ----Checking if the SST data, EEZ data, and depth data have the same CRS----
if(st_crs(sst) == st_crs(depth)){
  print("sst and depth CRS match!")
} else {
  warning("sst and depth CRS do not match.")
```

```
}


if(st_crs(sst) == st_crs(eez)){
  print("sst and eez CRS match!")
} else {
  warning("sst and eez CRS do not match.")
}


if(st_crs(eez) == st_crs(depth)){
  print("eez and depth CRS match!")
} else {
  warning("eez and depth CRS do not match.")
}
```

```
[1] "sst and depth CRS match!"
[1] "sst and eez CRS match!"
[1] "eez and depth CRS match!"
```

Process Data

```
# Mean SST from 2008-2012 in Kelvin
avg_sst_K <- mean(sst, na.rm = TRUE)

# Convert average from Kelvin to Celsius
avg_sst_C <- avg_sst_K - 273.15

# Crop depth to match SST raster
crop_depth <- crop(depth, ext(avg_sst_C))

# Use project() to get the CRS to match
crop_depth <- project(depth, avg_sst_C)

# Resample depth to the `sst` grid so both rasters share the same resolution,
origin, and extent
# `crop_depth` provides the bathymetry values to transfer
# `avg_sst_C` provides the grid, its temperature values are ignored
resampled_depth <- resample(
                            crop_depth,
                            avg_sst_C,
                            method = 'bilinear'
```

```
                            )
cat("After resampling to SST grid:\n")
cat("Resolutions match:", all(res(avg_sst_C) == res(resampled_depth)), "\n")
cat("Origins match:", all(round(origin(avg_sst_C), 8) ==
                            round(origin(resampled_depth), 8)), "\n")
cat("Extents match:", ext(avg_sst_C) == ext(resampled_depth), "\n")

# Combine SST and Depth into a raster stack
# This is a check that the rasters align, because otherwise c() would give
an error
temp_depth <- c(avg_sst_C, resampled_depth)
```

```
After resampling to SST grid:
Resolutions match: TRUE
Origins match: TRUE
Extents match: TRUE
```

Creating a function to look at the suitable areas within EEZs for select species

```
suitable_areas    <-    function(minimum_sst,    maximum_sst,    minimum_depth,
maximum_depth, species_name) {
    # Reclassification matrix for sst:
    # each row is (from, to, new value)
    # below the minimum_sst is unsuitable
    # between the minimum_sst and maximum_sst is suitable
    # above the maximum_sst is unsuitable
    sst_rcl <- matrix(c(-Inf, minimum_sst, 0,
                        minimum_sst, maximum_sst, 1,
                        maximum_sst, Inf, 0),
                      ncol = 3, byrow = TRUE)

    # Reclassification matrix for depth:
    # below the minimum_depth is unsuitable
    # between the minimum_depth and maximum_depth is suitable
    # above the maximum_depth is unsuitable
    depth_rcl <- matrix(c(-Inf, minimum_depth, 0,
                          minimum_depth, maximum_depth, 1,
                          maximum_depth, Inf, 0),
                        ncol = 3, byrow = TRUE)

    # Classify rasters based on RCL
```

```
    # Produces `avg_sst_C` raster with 0s and 1s based on suitability determined
by `sst_rcl`
    sst_suitable   <- classify(avg_sst_C, rcl = sst_rcl)
     # Produces `resampled_depth` raster with 0s and 1s based on suitability
determined by `depth_rcl`
    depth_suitable <- classify(resampled_depth, rcl = depth_rcl)

   # ----Find locations that satistfy the species SST and depth requirements----

    # Combine rasters for suitable SST zones with suitable depth zones
    sst_depth_zones <- c(sst_suitable, depth_suitable)

   # Multiplying 0s and 1s to find locations that meet both SST and depth criteria
   suitable_loc <- lapp(sst_depth_zones, fun = function(sst, depth) sst * depth)

    # ----Determine the most suitable EEZ----

    # Set unsuitable cells (0) to NA so only suitable cells remain
    suitable_loc[suitable_loc == 0] <- NA

    # Rasterize the EEZ polygons onto the suitability grid, so each cell holds
its EEZ name
    eez_rast <- rasterize(eez, suitable_loc, field = "rgn")

    # Keep only suitable cells that fall inside an EEZ
    suitable_eez <- mask(suitable_loc, eez_rast)

    # ----Find the area of grid cells----

    # Calculate the area of each cell in km^2
    # Cell area varies with latitude so it cannot be determined from resolution
alone
    cell_area <- cellSize(suitable_eez, unit = "km")

    # Area of each suitable cell (NA cells stay NA)
    suitable_area <- suitable_eez * cell_area

    # ----Find the total suitable area within each EEZ----

    # Sum the suitable area within each EEZ zone
     eez_suitable_area <- zonal(suitable_area, eez_rast, fun = "sum", na.rm =
TRUE) |>
         # rename the `lyr1` column that lists the total suitable area within
```

```
each EEZ to `area`
        rename(area = lyr1)

    # Create a tibble of suitable zones
    zones_tibble <- as_tibble(eez_suitable_area)

    # Joining suitable area to EEZ polygons
    eez_zones <- left_join(eez, zones_tibble, by = "rgn")

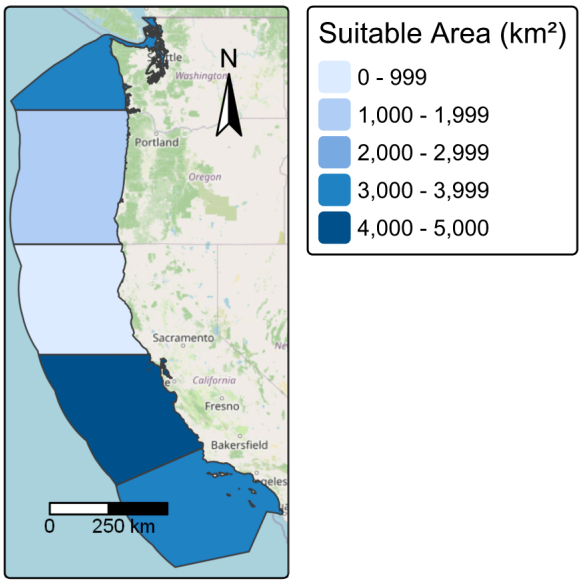
    # Create map
    tm_shape(eez_zones) +
        tm_basemap("OpenStreetMap") +
        tm_fill(
        fill = "area",
        fill.legend = tm_legend(title = "Suitable Area (km²)")
        ) +
         tm_scalebar(position = c("left", "bottom"), breaks = c(0, 250, 500),
text.size = 0.6) +
        tm_compass(position = c("right", "top")) +
        tmap_options(component.autoscale = FALSE) +
        tm_borders(lwd = .8) +
        tm_title(paste("Suitable", species_name, "Habitats by West Coast EEZ"),
size = 0.80)

}
```

Using the function to look at Oysters

```
suitable_areas(minimum_sst = 11, maximum_sst = 30, minimum_depth = -70,
maximum_depth = 0, species_name = "Oyster")
```

Suitable Oyster Habitats by West Coast EEZ



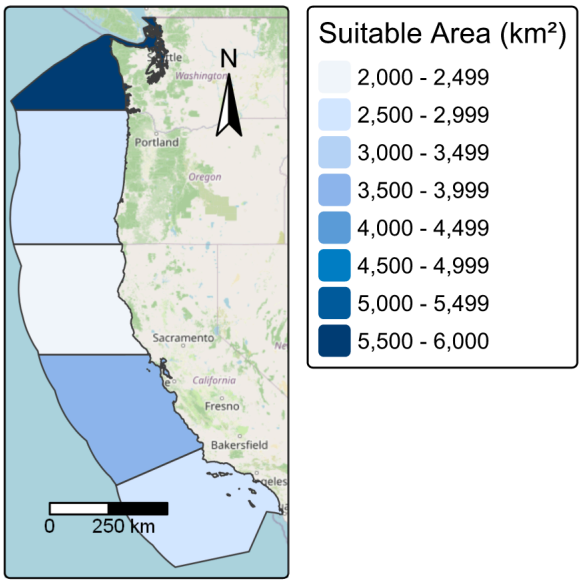
Interpretation of the map

The optimal growth conditions for oysters are between 11 degrees Celsius and 30 degrees Celsius and between sea level and 70 m below sea level. The map featured above shows the Central California EEZ has the most amount of suitable area for oyster growth (4,374 km²). On the other hand, the Northern California EEZ has the least amount of suitable area for oyster growth (324 km²).

Using the function to look at Red Rock Shrimp

```
suitable_areas(minimum_sst = 10, maximum_sst = 20, minimum_depth = -60,
maximum_depth = 0, species_name = "Red Rock Shrimp")
```

Suitable Red Rock Shrimp Habitats by West Coast EEZ



Interpretation of the map

The optimal growth conditions for red rock shrimp are between 10 degrees Celsius and 20 degrees Celsius and between sea level and 60 m below sea level. The map featured above shows the Washington EEZ has the most amount of suitable area for red rock shrimp growth (5,629 km²). On the other hand, the Northern California EEZ has the least amount of suitable area for oyster growth (2,311 km²).

Data Citations:

- Flanders Marine Institute (2023). Maritime Boundaries Geodatabase: Maritime Boundaries and Exclusive Economic Zones (200NM), version 12. Accessed on November 02, 2026 at https://www.marineregions.org/eez.php.
- GEBCO Bathymetric Compilation Group 2026 (2026). The GEBCO_2026 Grid - a continuous terrain model for oceans and land at 15 arc-second intervals. Accessed on November 02, 2026 at https://www.gebco.net/data-products/gridded-bathymetry-data#area.
- NOAA Coral Reef Watch (2026). NOAA Coral Reef Watch Version 3.1 Daily Global 5km Satellite Sea Surface Temperature Anomaly Product, 2008–2012. College Park, Maryland, USA: NOAA Coral Reef Watch. Accessed on November 02, 2026 at https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php.